CS8850: Advanced Machine Learning

Sergey Plis, Ph.D.

Fall 2026

E-mail:	splis@gsu.edu	Web:	iCollege
Office Hours:	Mon&Wed 14:00-14:30	Class Hours:	Mon&Wed 14:30-16:15
Office:	55 Park PLace, rm. 1821	Class Room:	55 Park PLace, rm. 1802

Note, most lecture slides and video recorded sessions in the tutorial style are available for your convenience at this page and in this Youtube playlist.

GSU Policy

The selling, sharing, publishing, presenting, or distributing of instructor-prepared course lecture notes, videos, audio recordings, or any other instructor-produced materials from any course for any commercial purpose is strictly prohibited unless explicit written permission is granted in advance by the course instructor. This includes posting any materials on websites such as Chegg, Course Hero, OneClass, Stuvia, StuDocu and other similar sites. Unauthorized sale or commercial distribution of such material is a violation of the instructor's intellectual property and the privacy rights of students attending the class, and is prohibited.

Teaching Assistant

Name:	Mansoor Ahmed
E-mail:	mahmed76@student.gsu.edu
Office Hours:	by appointment 4 hours a week
Location:	slack, online chat WebEx, Zoom, or similar

Course Description

Machine learning studies algorithms that build models from data for subsequent use in prediction, inference, and decision making tasks. Although an active field for the last 60 years, the current demand as well as trust in machine learning exploded as increasingly more data become available and the problems needed to be addressed become literally impossible to program directly. In this a dvanced course we will cover essential algorithms, concepts, and principles of machine learning. Along with the traditional exposition we will learn how these principles are currently being revisited thanks to the recent discoveries in the field.

Pre-requisites

- Programming in at least one programming language. Preferably in python, as homework will be given with expectation of python knowledge.
- Basic knowledge of linear algebra, optimization theory, probability, and statistics. The class is taught under assumption that you have taken some machine learning, statistics, calculus, probability, and optimization theory classes and are ready for advanced material.
- The class is taught under assumption that you are either far along in your graduate studies and/or are ready to act responsibly as a mature graduate student:
  - LaTeXtypesetting of all your material turned in for grading is assumed
  - Ability to code or learn to code in python is assumed.
  - Ability to decide and execute whether to typeset your work in LaTeXand turn in pdfs generated thereby or typeset everything including explanations, figures, and equations in a jupyter notebook is assumed.
  - Although I will try to cover every detail you may need for this class, ability to find and consult any additional material you may require is assumed.
  - Ability to initiate communication and ask questions when anything is unclear is also assumed.

  If you think you are not yet at the stage to be comfortable with above, the class may end up difficult for you for wrong reasons.

Reading List

I will use parts of the following text books, accompanied by mandatory research papers and optional reading material that I may recommend to supplement the lectures.

- Bishop CM. "Pattern recognition and machine learning." Springer; 2006
- Shalev-Shwartz S, Ben-David S. "Understanding machine learning: From theory to algorithms." Cambridge university press; 2014
- MacKay DJ "Information theory, inference and learning algorithms." Cambridge university press; 2003

- Goodfellow I, Bengio Y, Courville A. "Deep learning." MIT press; 2016
- Provost F, Fawcett T. "Data Science for Business: What you need to know about data mining and data-analytic thinking." 2013

Course Policy

I will detail the policy for this course below.

Communication

If you need to meet me in person, please use the office hours window or email me first to make an appointment. To minimize the disappointment of not getting attention, please avoid showing up at the instructor's office without a prior appointment. If it is a quick question please ask in slack preferably in a public channel where others can benefit from my answer as well or send a quick email.

Grading Policy

- **10%** of your grade will be determined by your homework grades.
- **30%** of your grade will be determined by a midterm during normal class hours.
- **30%** of your grade will be determined by the final exam during normal class hours.
- **30%** of your grade will be determined by the course project.

The scheme could change at the instructor's discretion, of which you will be notified in advance.

Below is the tentative schedule of the class lecture by lecture. Test column contains codes and sometimes explanations of the kind of test administered that day or homework due. 'HWi' - homework issued or due, 'E' - written timed exam, 'P' - project related deadlines.

	date	topic	tasks
1	Mon Aug 24	Introduction	
2	Wed Aug 26	Foundations of learning	Drop/Add deadline
3	Mon Aug 31	PAC learnability	HW1
4	Wed Sep 02	Linear learning models	
5	Mon Sep 07	Labor Day Holiday	Holiday
6	Wed Sep 09	Principal Component Analysis	Project ideas
7	Mon Sep 14	Curse of Dimensionality	HW2, HW1 due
8	Wed Sep 16	Bayesian Decision Theory	
9	Mon Sep 21	Parameter estimation: MLE	
10	Wed Sep 23	Parameter estimation: MAP & Naive Bayes	P (finalize teams)
11	Mon Sep 28	Logistic Regression	HW3, HW2 due
12	Wed Sep 30	Kernel Density Estimation	
13	Mon Oct 05	Support Vector Machines	
14	Wed Oct 07	Matrix Factorization	
15	Mon Oct 12	* Midterm	Exam, HW3 due
16	Wed Oct 14	k-means clustering	
17	Mon Oct 19	Invited: Intro to Causal Learning I	
18	Wed Oct 21	Invited: Intro to Causal Learning II	
19	Mon Oct 26	* Mid-point projects checkpoint	HW4, P
20	Wed Oct 28	Expectation Maximization	
21	Mon Nov 02	Stochastic Gradient Descent	
22	Wed Nov 04	Automatic Differentiation	
23	Mon Nov 09	Nonlinear embedding approaches	HW5, HW4 due
24	Wed Nov 11	Model comparison	
25	Mon Nov 16	Model Calibration	
26	Wed Nov 18	Convolutional Neural Networks	
27	Mon Nov 23	Thanksgiving Break	Holiday
28	Wed Nov 25	Thanksgiving Break	Holiday
29	Mon Nov 30	Word Embedding	
30	Wed Dec 02	Project Final Presentations	HW5 due, P
31	Mon Dec 07	Presentation spillover / Exam prep	Classes End
32	Wed Dec 09	* Final Exam	Exam
33	Wed Dec 16	Project Reports	due
34	Thu Dec 17	Grades due 5 p.m.	PAWS

Homework Assignments

All homework and extra credit submissions (if any) should be uploaded on iCOLLEGE dropbox into the relevant folder. The system also manages cut-off time, so 1 minute late is still late! Prepare accordingly. Regular completion of all assignments is critical to succeed in this course.

Assignments should be **TYPED, NOT HANDWRITTEN and/or SCANNED**; Submissions should be either in MS Office/OpenOffice or pdf formats.

If the TA cannot open and read it, or if you submit flawed document, the grade is zero by default. No excuses here! It is a student's responsibility to insure the file opens correctly. After you submit on iCOLLEGE, go and try to open/download it. Always make sure that you hit "submit" button,

and get a notification from iCOLLEGE that your submission went through.

Late Assignments

An assignment submitted past due will be given credit reduced by 2 for each past due day (counted from the official deadline). For example, a homework scored 96% but submitted one day late will only receive 48% for the grade; the same homework submitted 2 days late will only receive 24%. In general, all homework will be scored as score/2^n, where $n \in \{0, 1, ...\}$ is the number of days past due.

Missed Exams

Makeup exams are not given. Exceptions to this policy will be made only in case of a family or medical emergency, which must be clearly documented. **NO LAME EXCUSES, PLEASE!**

Academic Dishonesty Policy

Students are expected to submit their own work, to report only on what they completed/attended, and to conduct themselves in an honest manner. Students who do not follow these guidelines will be handled in a manner explained in the University Catalog. Please read the section on Academic Honesty and be honest in your submissions for this course.

One aspect of academic honesty is plagiarism. Plagiarism can include one or more of the following situations:

1. Copying information from another student's work or from other printed materials and submitting that work as your own.
2. Using other people's ideas, words, or data without properly documenting or acknowledging the outside source.
3. Overusing outside sources without incorporating your own ideas.

Any work containing plagiarized material will be graded as zero and cannot be made up. In some cases of plagiarism, students will receive a failing grade for the course. (See the University Catalog or the University website for additional information on academic honesty and plagiarism.) Take plagiarism seriously, in this and other classes. If you are not sure how to document something or what information must be documented, let your instructor know so that you can go over it together.

Students must work individually on the exam without any assistance from persons or things. Any student found to be cheating on an examination will receive a score of 0 for the exam, and possibly be given an F in the class. It is the student's responsibility to protect work from copying. No outside help is permitted. If a book or paper is used for the homework assignment, it must be referenced and not copied. Plagiarized work is determined solely by the instructor and teaching assistants and is graded solely at their discretion.

Closing of the University

Sometimes GSU closes because of weather conditions. To find out if the university is closed, go to www.gsu.edu or listen for an announcement on local TV or radio channels. If class is canceled, planned tests will take place during the next class session.

Students with Special Needs

Students who need special accommodations should arrange a meeting with the instructor as soon as possible. Bring a copy of your Student Accommodation Form to the meeting. If you do not have this form, make an appointment with the university's Office of Disability Services, 230 New Student Center, 404-413-1560.

Technology

GSU email

All students are **required** to activate and use their university e-mail account. Your instructor will send class updates and reminders via your GSU e-mail account. Please check your GSU e-mail regularly (daily) or have it forwarded to your personal account.

When emailing instructor or teaching assistants, you should do so from your students email account. Please also use meaningful subject line containing "[CS 8850]";

iCOLLEGE

You are encouraged to check iCOLLEGE often for announcements and useful links/other materials posted. You'll be using iCOLLEGE for submitting homeworks as well. You are encouraged to communicate with your classmates there, creating "discussions".

slack

You all will be required to join the class slack, as most of synchronous and weakly asynchronous communication will happen there. Please, use slack for communication with the instructor and other students in open channels, so everyone can benefit from answers and contribute to questions.

This syllabus represents a general plan for the course, deviations may be necessary